

Malek Dinari

 Malek-Dinari |  Malek Dinari |  dinari.malek1@gmail.com |  +216 53 828 817

SUMMARY

Computer Science Engineer (ENSI, 2025) with 13+ months of research experience in computer vision, deep learning, and AI applied to healthcare. Built end-to-end ML pipelines (PyTorch, TS-CAN) for contactless vital signs estimation, and full-stack AI applications (FastAPI, Next.js, local LLMs). Seeking an AI/ML Engineer or MLOps role to bring deep learning and production ML expertise to industry.

EDUCATION

National School of Computer Science (ENSI), University of Manouba 2021 – 2025

Computer Science Engineering Degree (graduated June 2025).

Coursework: Data Structures, Databases, Distributed Systems, Image Processing, AI, Statistics, Computational Complexity.

Preparatory Institute for Engineering Studies of Nabeul (IPEIN) 2019 – 2021

Mathematics–Physics track — Ranked 193 / 1779

EXPERIENCE

Researcher — CRISTAL Laboratory & cAIre (Healthcare AI) Sep 2025 – Mar 2026

- Designed a contactless blood pressure monitoring system using remote photoplethysmography (rPPG) to extract cardiovascular signals from standard video.
- Built an end-to-end deep learning pipeline around TS-CAN (PyTorch): multi-ROI patch decomposition, attention-guided rPPG signal recovery with auxiliary ridge-regression loss, and multi-region aggregation.
- Developed a real-time GUI for live BP monitoring and waveform visualization with side-by-side contact-based reference comparison.
- Presented a poster at **RFMI 2026** on ML approaches for vital signs estimation.

Research Intern — CRISTAL Laboratory, ENSI Summer 2025 (2 months)

- Studied and refined geodesic shape descriptors for 3D face analysis using statistical learning methods.
- Implemented mesh processing, geometric feature extraction, and evaluation of invariant shape descriptors.

Mobile Developer Intern — Welyne (IT Startup) Summer 2024 (3 months)

- Developed **WeeFizz**, a React Native size-recommendation app integrating QR scanning, user body-parameter collection, and photo capture for personalized clothing size suggestions.

PROJECTS

LearnLoop — AI-Powered Adaptive Study Platform (WIP) [GitHub](#)

- Full-stack platform that generates quizzes from uploaded documents using a local LLM (Ollama + Qwen).
- **Stack:** FastAPI backend (PDF/TXT extraction, chunking, embedding) + Next.js 14 / TypeScript frontend with real-time feedback and Socratic coaching chat.
- Operational features: multi-format quiz generation (MCQ, short-answer, T/F), instant LLM-graded answers with explanations. Roadmap: SSE streaming, PostgreSQL + Redis persistence, Docker Compose.

Bone Marrow Smear Segmentation [GitHub](#)

- Built an AI pipeline to classify and segment bone marrow cells for diagnostic support. Fine-tuned MobileNet for segmentation, then YOLOv8 for multi-class cell detection. Deployed with Gradio for interactive use.

Mini-GPT — Transformer from Scratch

[GitHub](#)

- Implemented a GPT-like autoregressive language model from scratch (bigram baseline → attention-based transformer) to deepen understanding of self-attention and generative architectures.

3D Facial Expression Recognition

[GitHub](#)

- Developed geodesic potential descriptors for 3D mesh-based facial expression recognition: level curve extraction, geodesic distance computation (Dijkstra), ICP mesh matching.

PUBLICATION

Poster — RFMI 2026 (IXème Atelier RFMI)

Jan 2026

Apprentissage machine pour l'estimation des paramètres vitaux

M. Dinari, E. Ghorbel, F. Ghorbel — CRISTAL Lab / GRIFT, Univ. de Manouba, in collaboration with cAIre.

CERTIFICATES

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- **Build Multimodal Generative AI Applications** — IBM, Mar 2026
 - **Advanced RAG with Vector Databases and Retrievers** — IBM, Mar 2026
 - **RAG Application Development (3-course series)** — IBM, Feb 2026
 - **Intermediate Docker** — DataCamp, Jan 2026
 - **Advanced Git** — DataCamp, Jan 2026
 - **Transformer Models with PyTorch** — DataCamp, Nov 2025

SKILLS

ML / AI	PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, NumPy/SciPy, signal processing (FFT, wavelets), YOLOv8
Languages	Python (primary), C/C++, JavaScript/TypeScript, SQL, Java
Full-Stack & Tools	FastAPI, Next.js, React Native, Angular, Spring Boot, Docker, Git, Streamlit, Gradio
Platforms	Linux (Ubuntu/WSL2), Kaggle, Google Colab, conda
Languages	English (fluent), French (proficient), Arabic (native)